

APPLICATIONS ASSIGNING SCORES WHEN MODELLING ORDINAL DATA

W.J. Conradie

Department of Statistics and Actuarial Science, University of Stellenbosch,
Private Bag X1, Matieland 7602, South Africa.

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Summary: It is not always obvious how to assign scores to the categories of ordinal variables when ordinal loglinear models are fitted to multidimensional contingency tables. In this paper, different scoring procedures are considered. In order to assess the practical implications of these different procedures, two well-known published data sets are analysed by fitting ordinal loglinear models. The purpose of this paper is to illustrate the different scoring procedures and to determine whether contradictory results are obtained. The danger of evaluating ordinal models by considering only goodness of fit, ignoring the relevance of the scores, will also be highlighted.

1. Introduction

When fitting nominal loglinear models to multidimensional contingency tables with one or more ordinal variables, the ordinal nature of any of these variables is ignored. Nominal models treat all classifications as nominal, in the sense that parameter estimates and test statistics are invariant with respect to orderings of categories. Thus not all the information in the data is used in the modelling of such contingency tables. This is a serious constraint since ordinal data are very common in practice.

This has led to the formulation of loglinear models for ordinal categorical models, i.e. loglinear models that exploit the ordinality of these ordinal variables. Ordinal models have association and interaction terms that reflect ordinal characteristics such as “monotone trend” or “stochastic ordering”. Advantages of these models over the nominal-scale models include the following:

1. Model parameters describe types of trends and are simpler to interpret than those in nominal models.
2. In some situations where nominal models are saturated it may be possible to find satisfactorily fitting, unsaturated ordinal models. This fact is illustrated very clearly in the analysis of table 3 (the second example in this paper).
3. Tests based on ordinal models have improved power for detecting certain types of association and interaction.

A problem with ordinal models is that they require **assigning of scores to the categories** and it is not always obvious how these scores should be assigned. Often the researcher may not want to assume equal spacings although the choice of equal-interval scores results in the simplest interpretation of the model. In many applications, the choice of scores will reflect assumed distances between categories of an underlying interval scale. When the choice of scores is not clear, it is informative to assign scores in a variety of “reasonable” ways. Conclusions may differ substantially depending on the choice of scores. For many variables no obvious choice of scores exists, yet parameter estimates and the goodness of fit of the models depend on that choice. Researchers are also often tempted to use the scoring procedure which leads to the best fit ignoring

the underlying rationale of the different scoring procedures. This can lead to some bias in the final conclusions.

In this paper, different scoring procedures proposed in the literature are considered. In order to assess their practical implications, two well-known data sets are analysed by means of fitting ordinal loglinear models, using different scoring procedures. The purpose of this paper is thus to determine whether the different scoring procedures lead to contradictory results and also to highlight the danger of evaluating ordinal models by considering only goodness of fit, ignoring the relevance of the scores.

2. Formulation of the problem – an example

Consider the following three-dimensional contingency table, of operation (O), hospital (H) and dumping severity (D) presented by Grizzle et. al. (1969) and discussed in Agresti (1990). The operations refer to treatments for duodenal ulcer patients and dumping severity describes a possible undesirable side effect of the operation. Both variables are ordinal and the different categories are described to the right of table 1. “Vagotomy” is the cutting of a nerve to reduce the flow of gastric juice, whilst “resection” refers to removal of part of the stomach. The data were collected from four hospitals.

Table 1

Cross-classification of Operation (O) Hospital (H) and Dumping Severity (D)					
O	H	D			TOTAL
		N	S	M	
A	1	23	7	2	32
	2	18	6	1	25
	3	8	6	3	17
	4	12	9	1	22
	TOTAL	61	28	7	96
B	1	23	10	5	38
	2	18	6	2	26
	3	12	4	4	20
	4	15	3	2	20
	TOTAL	68	23	13	104
C	1	20	13	5	38
	2	13	13	2	28
	3	11	6	2	19
	4	14	8	3	25
	TOTAL	58	40	12	110
D	1	24	10	6	40
	2	9	15	2	26
	3	7	7	4	18
	4	13	6	4	23
	TOTAL	53	38	16	107
					417

OPERATION

A Drainage and vagotomy

B 25% Resection and vagotomy

C 50% Resection and vagotomy

D 75% Resection and vagotomy

DUMPING

N Sweating

S Weakness

M Collapse of vascular or nervous system responses

This table can be collapsed over Hospital (**H**) since Dumping (**D**) is conditionally independent of Hospital (**H**) (cf. Agresti, 1990). The estimates of the standardized **OD**-association parameters suggest that an ordinal linear-by-linear model may be fitted on the collapsed (over **H**) table 2.

Table 2

CROSS-CLASSIFICATION OF OPERATION (O) AND DUMPING SEVERITY (D)				
OPERATION (O)	DUMPING			TOTAL
	N	S	M	
A	61	28	7	96
B	68	23	13	104
C	58	40	12	110
D	53	38	16	107
TOTAL	240	129	48	417

The **linear-by-linear** model follows as $\log m_{ik} = \mu + \lambda_i^O + \lambda_k^D + \beta u_i v_k$. The problem is now to assign the scores u_i and v_k sensibly. This matter has received considerable attention in the literature (c.f. Gross 1981, Agresti 1983, 1984, 1990, Graubhard & Korn 1987, Lipsitz 1992, Caudill 1993 and Randall 1993).

3. Scoring procedures: Linear-by-linear model

3.1 Equal-interval scores

Let $u_i = i$ and $v_k = k$ in table 2 and fit the linear-by-linear model $\log m_{ik} = \mu + \lambda_i^O + \lambda_k^D + \beta u_i v_k$. This model fits well with $G^2 = 4.59$, $df = 5$, $p = 0.468$, $\hat{\beta} = 0.1626$ ($ASE = 0.0656$). If the final category of dumping, i.e. moderate, is the baseline, the k th logit follows as:

$$\log \left(\frac{m_{ik}}{m_{i3}} \right) = \left(\lambda_k^D - \lambda_3^D \right) + \beta u_i (v_k - v_3) = \alpha_k + \beta i (k - 3)$$

with $\alpha_k = \lambda_k^D - \lambda_3^D$.

The odds of moderate ($k = 3$) instead of no ($k = 1$) or slight ($k = 2$) dumping, in terms of the model parameters, for operation i follow as:

$$\frac{m_{i3}}{m_{ik}} = \exp(-\alpha_k + \beta i (3 - k))$$

and for operation $(i + 1)$ as:

$$\begin{aligned} \frac{m_{i+1,3}}{m_{i+1,k}} &= \exp(-\alpha_k + \beta (i + 1) (3 - k)) \\ &= \exp(-\alpha_k + \beta i (3 - k)) \cdot \exp((3 - k) \beta) . \end{aligned}$$

Hence the relationship between the odds of moderate ($k = 3$) instead of no ($k = 1$) or slight ($k = 2$) dumping for operation $(i + 1)$ and operation (i) is given by:

$$\frac{m_{i+1,3}}{m_{i+1,k}} = \exp((3 - k) \beta) \cdot \frac{m_{i3}}{m_{ik}} \quad (1)$$

i.e.

$$\frac{m_{i+1,3}}{m_{i+1,1}} = \exp(2\beta) \cdot \frac{m_{i3}}{m_{i1}} \quad \text{and} \quad \frac{m_{i+1,3}}{m_{i+1,2}} = \exp(\beta) \cdot \frac{m_{i3}}{m_{i2}} .$$

These expressions give a very clear interpretation for β . The odds of moderate instead of no ($k = 1$) or slight ($k = 2$) dumping increase multiplicatively by $\exp((3 - k) \beta)$ for every unit increase in the operation score, i.e. for each additional 25% of stomach removal.

The fitted model implies that the estimated odds of moderate instead of slight ($k = 2$) dumping, are $\exp((3 - 2) \hat{\beta}) = \exp(0.1626) = 1.18$ times higher for each additional 25% of stomach removal and the estimated odds of moderate instead of no ($k = 1$) dumping are $\exp((3 - 1) \hat{\beta}) = \exp(0.3252) = 1.38$ times for each additional 25% of stomach removal. A 95% confidence interval for β is $0.1626 \pm 1.96 \cdot ASE = [0.03402; 0.291176]$ which gives the following approximate 95% confidence intervals for $\exp((3 - k) \beta)$:

$$Conf(\exp((3 - 2) \beta))_{0.95} = [I_L^{0.95}; I_U^{0.95}] = [1.0346; 1.338]$$

and

$$\text{Conf}(\exp((3-1)\beta))_{0.95} = [I_L^{0.95}; I_U^{0.95}] = [1.0704; 1.7902] .$$

The second interval implies, for example, that we have 95% confidence that the odds of moderate dumping instead of no dumping for operation $(i+1)$ are approximately between 1.07 and 1.79 times the odds of moderate dumping instead of no dumping for operation i .

3.2 Expert scores

Suppose the following scores reflect, from a medical point of view, the degree of seriousness of the operations and dumping severity: $u_1 = 1, u_2 = 2, u_3 = 4, u_4 = 7, v_1 = 1, v_2 = 3, v_3 = 9$. The linear-by-linear model $\log m_{ik} = \mu + \lambda_i^O + \lambda_k^D + \beta u_i v_k$ fits relatively well with $G^2 = 6.51, df = 5, p = 0.2594$, and $\hat{\beta} = 0.0177 (ASE = 00.00845)$. With these scores the linear-by-linear model leads to the following expressions for the odds of moderate ($k = 3$) instead of no ($k = 1$) or slight ($k = 2$) dumping, in terms of the model parameters for operation i :

$$\frac{m_{i3}}{m_{ik}} = \exp(-\alpha_k + \beta u_i (v_3 - v_k))$$

and for operation $(i+1)$ as:

$$\begin{aligned} \frac{m_{i+1,3}}{m_{i+1,k}} &= \exp(-\alpha_k + \beta u_{i+1} (v_3 - v_k)) \\ &= \exp(-\alpha_k + \beta u_i (v_3 - v_k) \cdot \exp((u_{i+1} - u_i)(v_3 - v_k)\beta)) . \end{aligned}$$

Hence the relationship between the odds of moderate instead of no ($k = 1$) or slight ($k = 2$) dumping for operation $(i+1)$ and operation (i) is given by:

$$\frac{m_{i+1,3}}{m_{i+1,k}} = \exp((u_{i+1} - u_i)(v_3 - v_k)\beta) \cdot \frac{m_{i3}}{m_{ik}} \quad (2)$$

The odds of moderate ($k = 3$) instead of no ($k = 1$) or slight ($k = 2$) dumping increase multiplicatively by $\exp((u_{i+1} - u_i)(v_3 - v_k)\beta)$ for every unit increase in the operation score, i.e. for each additional 25% of stomach removal.

The fitted model implies that the estimated odds of moderate ($k = 3$) instead of slight ($k = 2$) dumping, are $\exp((u_2 - u_1)(v_3 - v_2)\hat{\beta}) = \exp((2 - 1)(9 - 3)\hat{\beta}) = \exp(6(0.0177)) = 1.11$ times higher for operation B than operation A, $\exp(12(0.0177)) = 1.23$ times higher for operation C than operation B and $\exp(18(0.0177)) = 1.37$ times higher for operation D than operation C. The corresponding quantities for moderate ($k = 3$) instead of no ($k = 1$) dumping are 1.15, 1.33 and 1.53. A 95% confidence interval for β is $0.0177 \pm 1.96 \cdot ASE = [0.001138; 0.034262]$ which gives the following approximate 95% confidence intervals for $\exp((u_{i+1} - u_i)(v_3 - v_k)\beta)$:

$$\begin{aligned} Conf(\exp((u_2 - u_1)(v_3 - v_2)\beta))_{0.95} &= [1.006851; 1.228227] \\ Conf(\exp((u_3 - u_2)(v_3 - v_2)\beta))_{0.95} &= [1.013750; 1.508543] \\ Conf(\exp((u_4 - u_3)(v_3 - v_2)\beta))_{0.95} &= [1.020695; 1.852833] \\ Conf(\exp((u_2 - u_1)(v_3 - v_1)\beta))_{0.95} &= [1.009146; 1.315341] \\ Conf(\exp((u_3 - u_2)(v_3 - v_1)\beta))_{0.95} &= [1.018375; 1.730122] \\ Conf(\exp((u_4 - u_3)(v_3 - v_1)\beta))_{0.95} &= [1.027688; 2.275701] \end{aligned}$$

3.3 Midrank scores

Take the midranks as the scores for the i th category of operation and the k th category of dumping, i.e.

$$u_i = \frac{1}{n} \left(n_{1\bullet} + \dots + n_{i-1\bullet} + \frac{1}{2}(n_{i\bullet} + 1) \right)$$

and

$$v_k = \frac{1}{2} \left(n_{\bullet 1} + \dots + n_{\bullet k-1} + \frac{1}{2}(n_{\bullet k} + 1) \right)$$

where $n_{i\bullet} = \sum_{k=1}^K n_{ik}$ = the i th row total and $n_{\bullet k} = \sum_{i=1}^I n_{ik}$ = the k th column total and $n = \sum_{i=1}^I n_{i\bullet}$.

The calculated scores (midranks) are $u_1 = 0.1163$, $u_2 = 0.3561$, $u_3 = 0.6137$, $u_4 = 0.8729$, $v_1 = 0.2890$, $v_2 = 0.7314$, $v_3 = 0.9435$. These scores can be standardized where the standardized value of u_i is defined as:

$$u_i^* = \frac{u_i - \bar{u}}{s_u}, \quad i = 1, \dots, 4$$

with

$$\bar{u} = \frac{1}{n} \sum u_i \cdot n_{i\bullet} \quad \text{and} \quad s_u^2 = \frac{1}{n-1} \sum n_{i\bullet} (u_i - \bar{u})^2$$

which results in

$$\bar{u}^* = \frac{1}{n} \sum u_i^* \cdot n_{i\bullet} = 0 \quad \text{and} \quad s_{u^*}^2 = \frac{1}{n-1} \sum n_{i\bullet} (u_i^* - \bar{u}^*)^2 = 1.$$

The standardized scores follow as $u_1^* = -1.378$, $u_2^* = -0.52$, $u_3^* = 0.4016$, $u_4^* = 1.329$, and similarly for v_1 as $v_1^* = -0.8335$, $v_2^* = 0.9042$, $v_3^* = 1.7376$. With these scores the linear-by-linear model also fits relatively well with $G^2 = 7.28$, $df = 5$, $p = 0.2009$, and $\hat{\beta} = 0.0941$ ($ASE = 0.0498$). As in 3.2 the relationship between the odds of moderate instead of no ($k = 1$) or slight ($k = 2$) dumping for operation ($i + 1$) and operation (i) is given by:

$$\frac{m_{i+1,3}}{m_{i+1,k}} = \exp((u_{i+1}^* - u_i^*)(v_3^* - v_k^*)\beta) \cdot \frac{m_{i3}}{m_{ik}} \quad (3)$$

The fitted model implies that the estimated odds of moderate ($k = 3$) instead of slight ($k = 2$) dumping are $\exp(0.715057(0.0941)) = 1.06$ times higher for operation B than operation A, $\exp(0.9216(0.0941)) = 1.09$ times higher for operation C than operation B and $\exp(1.107589(0.0941)) = 1.11$ times higher for operation D than operation C. The corresponding quantities for moderate ($k = 3$) instead of no ($k = 1$) dumping are 1.23,

1.25, and 1.25. A 95% confidence interval for β is $0.0941 \pm 1.96 \cdot ASE = [-0.003508; 0.191708]$ which gives the following approximate 95% confidence intervals for $\exp((u_{i+1}^* - u_i^*)(v_3^* - v_k^*)\beta)$:

$$\begin{aligned} Conf(\exp((u_2^* - u_1^*)(v_3^* - v_2^*)\beta))_{0.95} &= [0.997495; 1.146922] \\ Conf(\exp((u_3^* - u_2^*)(v_3^* - v_2^*)\beta))_{0.95} &= [0.997309; 1.158636] \\ Conf(\exp((u_4^* - u_3^*)(v_3^* - v_2^*)\beta))_{0.95} &= [0.997292; 1.159710] \\ Conf(\exp((u_2^* - u_1^*)(v_3^* - v_1^*)\beta))_{0.95} &= [0.992291; 1.526395] \\ Conf(\exp((u_3^* - u_2^*)(v_3^* - v_1^*)\beta))_{0.95} &= [0.991722; 1.575003] \\ Conf(\exp((u_4^* - u_3^*)(v_3^* - v_1^*)\beta))_{0.95} &= [0.991670; 1.579512] \end{aligned}$$

3.4 Normal scores

The well-known normal scores for the categories of operation and dumping are obtained from the row and column totals respectively along the following lines:

The normal score for the i th category of operation is defined as:

$$u_i = \frac{1}{p_{i\bullet}} \int_{b_{i-1}}^{b_i} u\phi(u)du$$

where $p_{i\bullet} = \frac{n_{i\bullet}}{n}$, $b_i = \Phi^{-1}\left(\frac{1}{n} \sum_{r=1}^i n_{r\bullet}\right)$, $b_0 = -\infty$, $b_4 = \infty$, ϕ is the p.d.f. and Φ the c.d.f. of the standard $n(0, 1)$ distribution. The normal scores of the categories of dumping follow similarly.

The calculated normal scores are $u_1 = -1.3197$, $u_2 = -0.3794$, $u_3 = 0.2891$, $u_4 = 1.2555$, $v_1 = -0.6802$, $v_2 = 0.6386$, $v_3 = 1.6873$. The standardized scores follow as $u_1^* = -1.4204$, $u_2^* = -0.4082$, $u_3^* = 0.3112$, $u_4^* = 1.3512$, $v_1^* = -0.8019$, $v_2^* = 0.7522$, $v_3^* = 1.9879$.

The linear-by-linear model fits well with $G^2 = 4.61$, $df = 5$, $p = 0.465$, and $\hat{\beta} = 0.1239$ ($ASE = 0.0500$). The relationship between the odds of moderate instead of no ($k = 1$) or slight ($k = 2$) dumping for operation ($i + 1$) and operation (i) is given by expression (3). The fitted model

implies that the estimated odds of moderate instead of slight dumping are $\exp(1.2506(0.1239)) = 1.1676$ times higher for operation B than operation A, $\exp(0.8891(0.1239)) = 1.1165$ times higher for operation C than operation B and $\exp(1.2852(0.1239)) = 1.1726$ times higher for operation D than operation C. The corresponding quantities for moderate instead of no dumping are 1.4188, 1.2823 and 1.4326. A 95% confidence interval for β is $0.1239 \pm 1.96 \cdot ASE = [0.0259; 0.2219]$ which gives the following approximate 95% confidence intervals for $\exp((u_{i+1}^* - u_i^*)(v_3^* - v_k^*)\beta)$:

$$\begin{aligned} \text{Conf}(\exp((u_2^* - u_1^*)(v_3^* - v_2^*)\beta))_{0.95} &= [1.032926; 1.319888] \\ \text{Conf}(\exp((u_3^* - u_2^*)(v_3^* - v_2^*)\beta))_{0.95} &= [1.003109; 1.026954] \\ \text{Conf}(\exp((u_4^* - u_3^*)(v_3^* - v_2^*)\beta))_{0.95} &= [1.054645; 1.577487] \\ \text{Conf}(\exp((u_2^* - u_1^*)(v_3^* - v_1^*)\beta))_{0.95} &= [1.075878; 1.871255] \\ \text{Conf}(\exp((u_3^* - u_2^*)(v_3^* - v_1^*)\beta))_{0.95} &= [1.007033; 1.061888] \\ \text{Conf}(\exp((u_4^* - u_3^*)(v_3^* - v_1^*)\beta))_{0.95} &= [1.127630; 2.798601] \end{aligned}$$

4. Evaluating different scoring procedures

The results of using different scoring procedures in fitting the linear-by-linear model to table 2 are studied by means of the estimated multiplicative constant in the relationship between the odds of moderate instead of no or slight dumping for operation $(i+1)$ and (i) , i.e. the quantity

$$\exp((u_{i+1} - u_i)(v_3 - v_k)\beta) \quad (4)$$

in expressions (1), (2) and (3) as well as their approximated 95% intervals.

The different scoring procedures all lead to reasonable and in some cases very good fits of the linear-by-linear model to table 2. If the goodness of fit statistic G^2 and its corresponding p -value are used as criteria for deciding on the most appropriate scoring procedure one might prefer the equidistant or normal procedures. In doing so the different interpretations as reflected

by the multiplicative factor in expression (4) are ignored. For example, for normal scores the fitted model implies that the estimated odds of moderate instead of no dumping are 1.42 times higher for operation B than operation A, while for the expert scores model the estimated odds of moderate instead of no dumping are 1.15 times higher for operation B than operation A; for expert scores the estimated odds of moderate instead of no dumping are 1.53 times higher for operation D than operation C, while for the midrank scores model this quantity is 1.25. Although the differences between these quantities for different scoring procedures are not everywhere substantial, it can lead to different interpretations if only the point estimates of the multiplicative factor are used as illustrated above. It is also very informative to look at the confidence intervals of the multiplicative factor. It is important to note that the corresponding intervals overlap for example for u_2 and $k = 1$ the intervals are [1.07; 1.79], [1.009; 1.315], [0.993; 1.526], and [1.076; 1.871] respectively. If the overlapping of the confidence intervals are taken as criteria it appears as if using different scoring procedures will not lead to significantly different interpretations of the fitted model. It is thus clear that there is no straightforward answer to the question of how sensitive ordinal analysis is to different scoring procedures. As Agresti (1990) puts it: "When there is no natural set of scores it is wise to conduct a sensitivity analysis, i.e. assign scores a variety of ways that seems reasonable to check whether substantive conclusions depend on the choice". If at all possible, expert scores based on sound practical insight and arguments should be used when ordinal models and the resulting assigning of scores are in question. These scores should correspond with the true unknown numerical scale that underlie the ordered classification.

5. Second example

The following contingency table from DiFrancisco and Critelman (1984) refers to the following of politics by different nations given different levels of education.

Table 3

THE FOLLOWING OF POLITICS BY DIFFERENT NATIONS GIVEN DIFFERENT LEVELS OF EDUCATION				
COUNTRY(C)	EDUCATION(E)	FOLLOW POLITICS(P)		
		YES	NO	TOTAL
USSR	Primary	94	84	178
	Secondary	318	120	438
	College	473	72	545
	Total	885	276	1161
USA	Primary	227	112	339
	Secondary	371	71	442
	College	180	8	188
	Total	778	191	969
UK	Primary	356	114	500
	Secondary	256	76	332
	College	22	2	24
	Total	634	222	856
ITALY	Primary	166	526	692
	Secondary	142	103	245
	College	47	7	54
	Total	355	636	991
MEXICO	Primary	447	430	877
	Secondary	78	25	103
	College	22	2	24
	Total	547	457	1004

A standard nominal log-linear analysis of this table reveals that the only model that fits the data is the saturated model, i.e.

$$\log m_{ijk} = \mu + \lambda_i^C + \lambda_j^E + \lambda_k^P + \lambda_{ij}^{CE} + \lambda_{jk}^{EP} + \lambda_{ik}^{CP} + \lambda_{ijk}^{CEP}.$$

If the ordinality of education is taken into account by assigning scores $\{v_j\}$ to its categories the following unsaturated mixed ordinal model can be considered:

$$\log m_{ijk} = \mu + \lambda_i^C + \lambda_j^E + \lambda_k^P + \lambda_{ij}^{CE} + v_j w_k \delta + w_k \phi_i + v_j w_k \beta_i$$

with $w_1 = 2$ when politics are followed regularly and $w_2 = 1$ otherwise.

The logit for the country (i) and education category (j) follows as:

$$\begin{aligned} \log(m_{ij1}/m_{ij2}) &= (\lambda_1^C - \lambda_2^P) + v_j \delta (w_1 - w_2) + \phi_i (w_1 - w_2) \\ &\quad + \beta_i v_j (w_1 - w_2) \\ &= \alpha + v_j \delta + \phi_i + v_j \beta_i \\ &= \alpha + \phi_i + v_j (\delta + \beta_i) \\ &= \phi_i^* + v_j \beta_i^* \end{aligned}$$

with $\sum_{i=1}^5 \phi_i = \sum_{i=1}^5 \beta_i = 0$, $\phi_i^* = \alpha + \phi_i$ and $\beta_i^* = \delta + \beta_i$.

The conditional cross product ratio of education level $j+1$ and j given country i is given by

$$\begin{aligned} \theta_{(i)j1} &= \frac{m_{i,j+1,1}/m_{i,j+1,2}}{m_{ij1}/m_{ij2}} \\ &= \exp(v_{j+1}(\delta + \beta_i) - v_j(\delta + \beta_i)) \\ &= \exp((v_{j+1} - v_j)(\beta_i^*)). \end{aligned}$$

The ratio of conditional cross product ratios of country r to country t is given by

$$\begin{aligned} \theta_{(rt)j1} &= \theta_{(r)j1} / \theta_{(t)j1} \\ &= \exp((\beta_r^* - \beta_t^*)(v_{j+1} - v_j)) \end{aligned}$$

which leads to

$$\theta_{(r)j1} = \exp((\beta_r^* - \beta_t^*)(v_{j+1} - v_j)) \cdot \theta_{(t)j1}. \quad (5)$$

The odds of following politics against not following it for education level $j+1$ relative to the odds for education level j in country r are

$\exp((\beta_r^* - \beta_t^*)(v_{j+1} - v_j))$ times the corresponding ratio in country t . If $\exp((\beta_r^* - \beta_t^*)(v_{j+1} - v_j)) > 1$ it means that the ratio of the odds of following politics instead of not following politics for education level $j + 1$ to the odds for education level j is larger in country r than in country t . In country r the interest in politics therefore increases more rapidly with an increase in education than in country t . Consequently, to evaluate the effect of the different scoring procedures on the practical interpretations, the estimated values of $\exp((\beta_r^* - \beta_t^*)(v_{j+1} - v_j))$ need to be calculated and further investigated. This can easily be done by using the estimates $\hat{\beta}_i^*$.

The different procedures of assigning scores to the categories of education lead to the following fits. For the expert scores the cumulative number of years at the end of each education phase is taken as the scores, i.e. $v_j = 7$, $v_2 = 12$ and $v_3 = 16$.

Table 4

SUMMARY OF MODEL FITS			
PROCEDURE	G^2	df	p-value
Equidistant	2.88	5	0.7181
Expert	6.21	5	0.2859
Normal	3.62	5	0.6046
Midrank	14.22	5	0.0143

Approximate confidence intervals for $\exp(\widehat{\beta}_r^* - \widehat{\beta}_t^*)(v_{j+1} - v_j)$ follow as

$$\exp(\widehat{\beta}_r^* - \widehat{\beta}_t^* \pm 1.96 \cdot \text{ASE}(\widehat{\beta}_r^* - \widehat{\beta}_t^*)(v_{j+1} - v_j))$$

where $\text{ASE}(\widehat{\beta}_r^* - \widehat{\beta}_t^*)$ is easily calculated from the variance-covariance matrix of the estimated parameters.

If the goodness of statistic G^2 and its corresponding p -value are used for deciding on the most appropriate scoring procedure, one might prefer the

equidistant, normal or expert scoring procedures for table 3. It is, however, important to take note of the different interpretations which arise from the different scoring procedures. This can be done by comparing the estimated values of $\exp((\beta_r^* - \beta_t^*)(v_{j+1} - v_j))$ for two countries under different scoring procedures. For example, comparing USSR with UK, a value of 1.6436 is obtained when equidistant scores are used while values of 0.796 and 0.8178 are obtained when standardized normal scores are used; comparing UK with ITALY, a value of 0.3316 is obtained when equidistant scores are used while values of 0.3147 and 0.3966 are obtained when expert scores are used and 0.7058 and 0.7355 when standardized normal scores are used.

A number of approximate confidence intervals for USSR→UK and USA→UK when standardized normal scores are used, do not overlap with the corresponding intervals when equal-interval scores and expert scores are used. This implies significant differences in the practical interpretation of the fitted models.

6. Concluding remarks

The different scoring procedures all lead to reasonable and in some cases very good fits of the fitted models in the two examples discussed in this paper. If the goodness of fit statistic G^2 and its corresponding p -value are used as the only criteria for deciding on the most appropriate scoring procedure, the different interpretations, as reflected by the multiplicative factor in expressions (1), (2) and (3) as well as the multiplicative factor in expression (5), are ignored. Although the differences between these quantities for different scoring procedures are not everywhere substantial, it can lead to different

interpretations as pointed out in the discussion of the examples. Practical aspects will, however, determine if these differences are substantial.

Assigning expert scores should always be attempted first since an expert in the field of origin of the data is in the best position to estimate the differences between the categories of ordinal categorical variables. For example, the medical practitioner is best qualified to assign scores to the different operations and dumping severity in table 1 and the educator is in the best position to assign scores to the different levels of education in table 3.

In conclusion, the analysis of the two contingency tables in this paper showed that using different scoring procedures when fitting ordinal models may all lead to good fits but can result in different interpretations of the data. Researchers should thus be very cautious when evaluating only goodness of fit in deciding on the most proper scoring procedure. The most important fact to emphasise is thus that a good fit does not necessarily imply a "correct" fit. The scoring procedure has a significant effect on the practical interpretation of the fitted model. If at all possible, expert scores based on sound practical insight and arguments should be used when ordinal models and the resulting assigning of scores are at stake. These scores should correspond with the true unknown numerical scale that underlies the ordered classification.

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